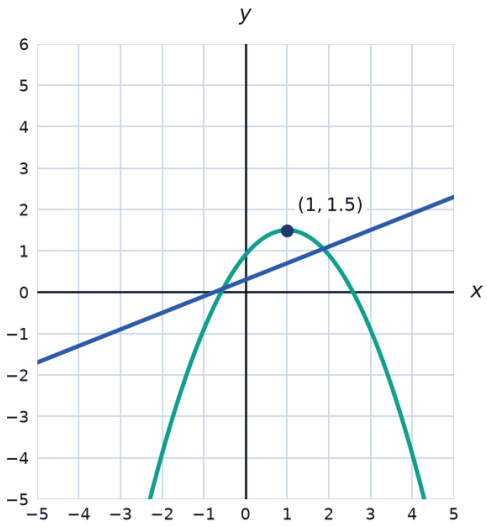
The Chain Rule & Directional Derivatives

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the directional derivative of $f(x, y) = x^2y - 3y^2$ at the point $P(1, 2)$ in the direction of the vector $\mathbf{v} = \langle 3, 4 \rangle$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For $f(x, y) = -12x - 35y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.



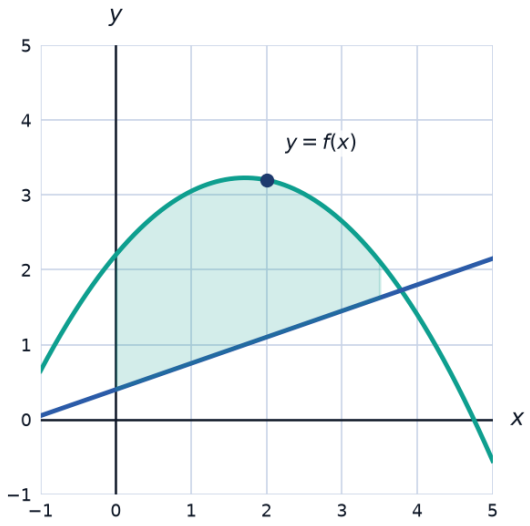
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Prove that the 3D vector field $\mathbf{F} = \langle 2xy + z^2,\ x^2 + 2yz,\ 2xz + y^2 \rangle$ is conservative on $\mathbb{R}^3$, construct a potential function f satisfying $ablaf = \mathbf{F}$, and evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0, 0)$ to $(1, 2, 3)$.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for the vector field $\mathbf{F} = \langle y, -x, z \rangle$ along the circular helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \le t \le 2\pi$.



5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

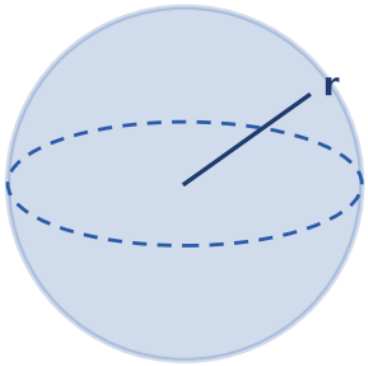
Consider the vector field $\mathbf{F}(x, y) = \langle 2x + y, x - 3y \rangle$. What is the value of the vector field at the point $(3, -1)$?

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Maximize $f(x, y) = 5xy$ subject to $x + y = -10$. Find the maximum value.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the total mass and the z-coordinate of the center of mass of the upper solid hemisphere $x^2 + y^2 + z^2 \le R^2$, $z \ge 0$, whose density is proportional to the distance from the origin: $\delta(\rho, \phi, \theta) = k\rho$ for constant $k > 0$.

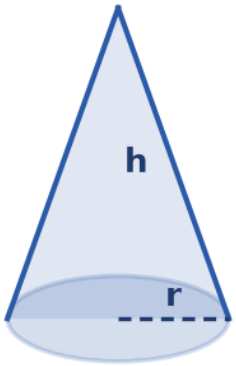


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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

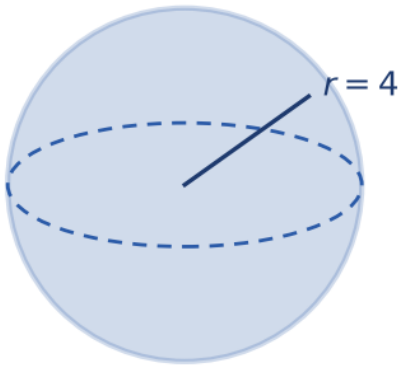
8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Compute the moment of inertia I_z about the z-axis of a solid homogeneous right circular cone of base radius R, height h, and uniform total mass M.



9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

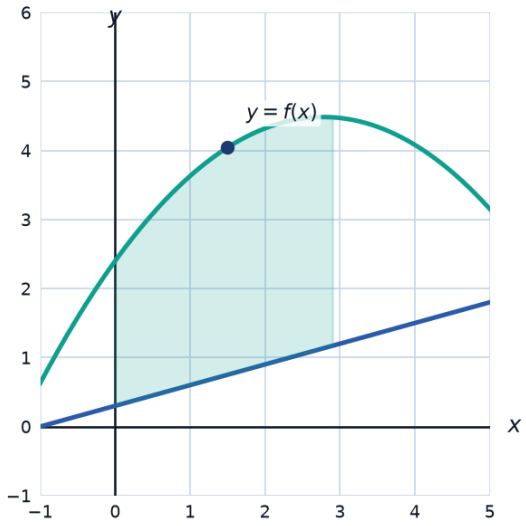
Find the volume and the z-coordinate of the centroid $\bar{z}$ of the solid region bounded below by the xy-plane $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$, assuming uniform density $\delta_0 = 1$.



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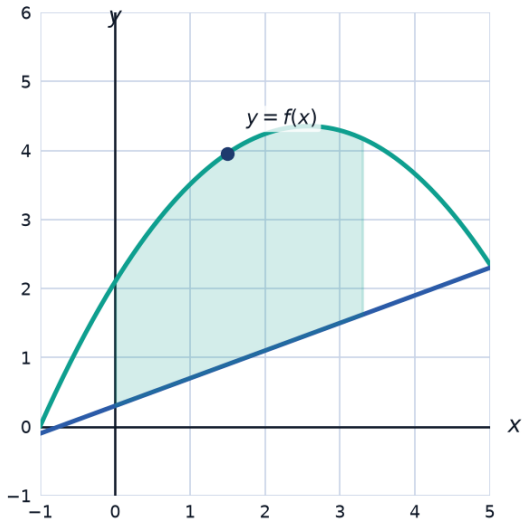
10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the Cartesian triple integral $\iiint_B xyz^2\,dV$ over the rectangular box $B = [0,1] \times [-1,2] \times [0,3]$.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

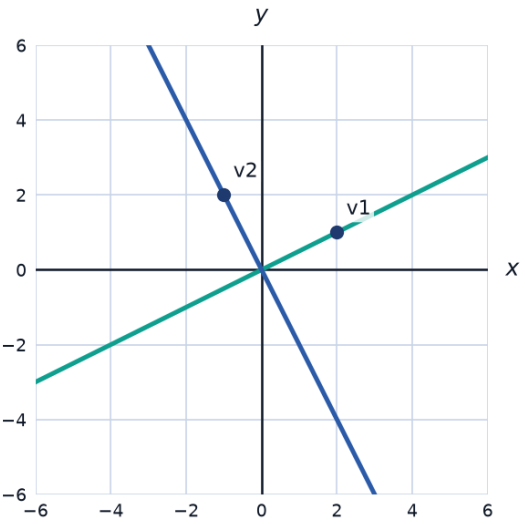
Evaluate the double integral $\iint_R x\cos(x+y)\,dA$ over the rectangular region $R = [0,\pi] \times [0,\pi/2]$.



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12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Classify the critical points $(0, 0)$ and $(1, 1)$ of $f(x, y) = x^3 + y^3 - 3xy$ using the Second Derivative Test and Hessian matrix.



13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the surface area of the paraboloid cap $z = x^2 + y^2$ that lies below the horizontal plane $z = 4$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Prove that the limit $\lim_{(x, y)\to(0, 0)} \frac{x^2y}{x^4 + y^2}$ does not exist by using the Two-Path Test, comparing straight-line approaches with parabolic approaches.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

A solid region E is defined by the inequalities $x^2 + y^2 \le 16$, $0 \le z \le 5$. The volume of this region can be computed using a triple integral. In cylindrical coordinates, the region E is described by $0 \le r \le R$, $0 \le \theta \le 2\pi$, and $0 \le z \le H$. What is the value of R?

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Find the directional derivative of $f(x, y, z) = x^2y + yz^2$ at $P(1, 2, -1)$ in the direction of $\mathbf{v} = \langle 2, -1, 2 \rangle$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Double Integrals

Let $z = x^2y - y^3$, where $x(t) = \cos(2t)$ and $y(t) = \sin(2t)$. Use the multivariable chain rule to find $\frac{dz}{dt}$ when $t = 0$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

[The Multivariable Chain Rule, Directional Derivatives & Gradient Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Given the multivariable function $f(x, y) = x^3y^2 - 4xy^3 + 5x^2 - 7y + 2$, find the first-order partial derivatives $f_x(x, y)$ and $f_y(x, y)$, and evaluate $f_x(1, 2)$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Double Integrals

Find an equation of the tangent plane to the surface $z = x^2 + 3y^2 - 2x + 4y$ at the point $P(2, 1, 9)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $z = f(x, y)$ where $x = g(t)$ and $y = h(t)$, what is $\frac{dz}{dt}$?

a. $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$

b. $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dy}{dt} + \frac{\partial z}{\partial y}\frac{dx}{dt}$

c. $\frac{dz}{dt} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$

d. $\frac{dz}{dt} = \frac{dx}{dt}\frac{dy}{dt}$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the gradient vector $\nabla f(x, y)$ of a scalar function $f(x, y)$?

a. $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$

b. $\nabla f = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$

c. $\nabla f = \left[\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}\right]$

d. $\nabla f = \sqrt{f_x^2 + f_y^2}$

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Compute the gradient $\nabla f(2, -1)$ for $f(x, y) = 3x^2y + 2y^2$.

a. $[-12, 8]$

b. $[12, -8]$

c. $[-12, -4]$

d. $[6, 4]$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the geometric relationship between the gradient $\nabla f(x_0, y_0)$ and the level curve $f(x, y) = c$ passing through (x_0, y_0)?

a. $\nabla f(x_0, y_0)$ is perpendicular (orthogonal) to the tangent line of the level curve

b. $\nabla f(x_0, y_0)$ is parallel to the tangent line

c. $\nabla f(x_0, y_0)$ has zero dot product with the normal vector

d. $\nabla f(x_0, y_0)$ points along the curve

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the formula for the directional derivative $D_{\mathbf{u}}f$ in the direction of a unit vector $\mathbf{u}$?

a. $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$

b. $D_{\mathbf{u}}f = \nabla f \times \mathbf{u}$

c. $D_{\mathbf{u}}f = ||\nabla f|| ||\mathbf{u}||$

d. $D_{\mathbf{u}}f = \frac{\nabla f \cdot \mathbf{u}}{||\nabla f||}$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Why must $\mathbf{u}$ be normalized to a unit vector before computing $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$?

a. To ensure the derivative measures rate of change per unit of geometric distance

b. Because dot products cannot be computed with non-unit vectors

c. To avoid division by zero

d. Because the gradient has length 1

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In which direction does a differentiable function f increase most rapidly at a point?

a. In the direction of the gradient vector ∇f

b. In the direction opposite to ∇f (i.e. $-\nabla f$)

c. Orthogonal to the gradient vector

d. Along the tangent to the level curve

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the maximum value of the directional derivative $D_{\mathbf{u}}f$ at a point?

a. $||\nabla f||$

b. $||\nabla f||^2$

c. 0

d. 1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In which direction is the rate of change of f equal to zero?

a. In any direction tangent to the level curve (orthogonal to ∇f)

b. In the direction of $-\nabla f$

c. In the direction of ∇f

d. Nowhere, a differentiable function always changes

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For $z = -3xy$ with $x = t^2$, $y = t$, find dz/dt at $t = 3$ using the chain rule.

a. -27

b. 81

c. -81

d. -78

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Chain rule with two paths: $z = f(x, y)$, $x = r\cos\theta$, $y = r\sin\theta$. $\partial z/\partial r =$

a. $f_x\cos\theta + f_y\sin\theta$

b. f_r

c. $f_x r + f_y r$

d. $f_x\sin\theta - f_y\cos\theta$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the vector field $\mathbf{F}(x, y) = \langle 2x, 3y\rangle$. Which of the following options best represents the vector field at the point (1, 1)?

a. $\langle 2, 3\rangle$

b. $\langle 1, 1\rangle$

c. $\langle 4, 6\rangle$

d. $\langle 2, 6\rangle$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the region in the first octant bounded by the coordinate planes and the plane $x + y + z = 4$. If the density of the region is given by $\rho(x, y, z) = 3xy$, set up the triple integral for the mass of this region in Cartesian coordinates.

a. $\int_0^4 \int_0^{4-x} \int_0^{4-x-y} 3xy\, dz\, dy\, dx$

b. $\int_0^4 \int_0^{4-y} \int_0^{4-y-z} 3xy\, dx\, dz\, dy$

c. $\int_0^4 \int_0^{4-x} \int_0^{4-x-y} 3\, dz\, dy\, dx$

d. $\int_0^4 \int_0^{4-z} \int_0^{4-z-y} 3xy\, dx\, dy\, dz$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.

a. $\sqrt{14}$

b. $2\sqrt{14}$

c. $3\sqrt{14}$

d. $4\sqrt{14}$

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)
Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.

a. 2

b. 4

c. 6

d. 0

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
Consider the vector field $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. Evaluate the vector field at the point $(2, 4)$.

a. $\langle 6, 8 \rangle$

b. $\langle 8, 6 \rangle$

c. $\langle 5, 6 \rangle$

d. $\langle 6, 5 \rangle$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle 0, 0, z \rangle$ across the upper hemisphere $x^2 + y^2 + z^2 = 1,\ z \geq 0$, with outward normal.

a. $\frac{2\pi}{3}$

b. $\frac{4\pi}{3}$

c. $-\frac{2\pi}{3}$

d. $\frac{\pi}{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives
Compute $\iint_D 9(x^2 + y^2)\, dA$ where D is the disk $x^2 + y^2 \leq 49$.

a. 2058π

b. 14406π

c. 21609π/4

d. 21609π/2

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers
Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y^2, 2x \rangle$ along the segment from $(-2,\ -2)$ to $(-2,\ 3)$.

a. 20

b. -19

c. -40

d. -20

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Double Integrals

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -2y, 2x, y \rangle$ around the circle $x^2 + y^2 = 1$ in the plane $z = 0$, oriented counterclockwise seen from above.

a. -4π

b. 2π

c. 8π

d. 4π

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -y, x, 0 \rangle$ around the boundary of the square $[0, 2] \times [0, 2]$ in the plane $z = 0$, oriented counterclockwise seen from above.

a. 8

b. -8

c. 4

d. 16

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle xy, x \rangle$ along the parabola $y = x^2$ from $(0,\ 0)$ to $(1,\ 1)$.

a. $-\frac{11}{12}$

b. $\frac{11}{6}$

c. $\frac{23}{12}$

d. $\frac{11}{12}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Double Integrals

Which vector field is conservative on all of R^2?

a. $\langle e^y, xe^y \rangle$

b. $\langle xy, x \rangle$

c. $\langle x + y,\ -x + y \rangle$

d. $\langle x^2 y, xy^2 \rangle$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Evaluate $\iiint_B (x^2 + y^2 + z^2)\, dV$ where B is the ball $x^2 + y^2 + z^2 \le 4$.

a. $\frac{128\pi}{5}$

b. $\frac{128\pi}{3}$

c. $\frac{64\pi}{5}$

d. $\frac{256\pi}{5}$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

$\mathbf{F} = \langle ye^x, e^x \rangle$ is conservative. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(2,\ 0)$ to $(1,\ -1)$.

a. e

b. $-e$

c. $1 - e$

d. $-2e$

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